CultiVar-17: One-Shot Digit Recognition with a Culturally-Motivated Style Taxonomy

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***Abstract*—We introduce CultiVar-17, a compact dataset of 17 hand-drawn digit templates that captures stylistic variation motivated by regional and educational handwriting conventions—crossed vs. uncrossed zero, European vs. American seven, serif vs. plain one, and more. Using only these 17 templates as supervision, we train a CNN and a prototype-embedding model via synthetic augmentation and evaluate transfer to the MNIST test set. Our best configuration—a prototype-embedding model with full augmentation—achieves $77.46 \pm 2.39\%$ MNIST accuracy from 17 examples, reaching 78% of supervised CNN performance. An ablation study shows that elastic distortion and stroke-width perturbation contribute independently and additively. All code and data are publicly available for reproducibility.**

***Index Terms*—one-shot learning, digit recognition, handwriting style, data augmentation, prototype networks, MNIST**

I. Introduction

Learning from very few examples is a longstanding challenge in machine learning [1]–[3]. Handwritten digit recognition has served as a foundational benchmark since LeCun et al. [4] established MNIST, yet standard pipelines assume thousands of labelled samples per digit class.

This work explores a strict one-shot regime: we build a complete recognition pipeline from exactly **one hand-drawn template per class**, structured around a culturally-aware 17-class taxonomy. Our contributions are threefold:

1) **CultiVar-17**—a 17-template dataset encoding documented regional writing variants, distributed as a reproducible `.npy` artifact.
2) **A one-shot augmentation pipeline**—elastic distortion, stroke-width perturbation, and affine noise applied to templates, generating diverse training batches from a single source.
3) **An empirical study**—comparing nearest-template, CNN, and prototype-embedding approaches with ablation over augmentation components, achieving 77.46% MNIST accuracy from 17 examples.

II. Related Work

A. One-Shot and Few-Shot Learning

Koch et al. [1] proposed Siamese Networks for one-shot image recognition, learning a similarity metric between pairs of examples. Vinyals et al. [2] introduced Matching Networks, which use attention-weighted nearest-neighbour classification in an embedding space conditioned on the support set. Snell et al. [3] proposed Prototypical Networks, classifying by computing Euclidean distances to class prototype embeddings—the architecture we adopt here. Finn et al. [5] introduced MAML, a gradient-based meta-learning approach that learns an initialisation enabling fast adaptation with few gradient steps.

B. Handwriting Datasets and Cultural Variation

Lake et al. [6] introduced Omniglot, a 1,623-character handwriting dataset covering 50 alphabets from diverse writing systems, which became the canonical few-shot classification benchmark. CultiVar-17 is complementary in scope: rather than cross-alphabet generalisation, it targets intra-digit stylistic variation driven by documented regional conventions. LeCun et al. [4] established MNIST as the standard digit recognition benchmark; we use MNIST as the transfer target throughout.

C. Data Augmentation for Handwriting

Simard et al. [7] demonstrated that elastic distortions—smooth random displacement fields applied to handwritten characters—substantially improve CNN generalisation on MNIST. We apply the same elastic distortion technique as a core augmentation component, combined with stroke-width perturbation. Our ablation (Section VI-B) confirms their independent and additive contributions.

III. CultiVar-17 Dataset

A. Cultural Motivation

Handwriting style reflects regional education systems and typographic conventions. We document seven split points in standard digit classes, shown in Table I. Digits 3, 5, 6, and 8 have no variant split, yielding $6 \times 2 + 1 \times 3 + 4 = 17$ classes in total.

TABLE I
CULTURAL MOTIVATION FOR CULTIVAR-17 VARIANT SPLITS.

Digit	Var. A	Var. B	Driver
0	uncrossed	crossed	zero-slash convention
1	plain vert.	serif / angled	American vs. European
2	looped base	angular base	cursive vs. print
4	open top	closed top (+ cursive)	printing vs. cursive
7	uncrossed	crossed	American vs. European
9	looped tail	straight tail	print vs. cursive

B. Template Extraction

Source: a single composite image generated with an AI image tool and manually verified for legibility and distinctiveness. Procedure: isolate the central strip; split into 17 equal horizontal slots; foreground-tight crop and centre each digit onto a 28×28 canvas. Figure 1 shows all 17 templates.

Output: `train_images.npy` (17 × 28 × 28 uint8) and `train_labels17.npy` (labels 0–16).

C. Class-to-Digit Mapping

For MNIST evaluation, 17 style classes are collapsed to 10 canonical digits via a fixed surjective map (`src/variants17/label_schema.py`). Each variant maps to its base digit (e.g., `0_variant_a` and `0_variant_b` → 0).

IV. Method

A. Augmentation Pipeline

Given one template per class, we generate training batches via stochastic augmentation:

- **Elastic distortion** [7]: random displacement field ($\sigma = 4$, $\alpha \in [15, 32]$), interpolated with Gaussian smoothing.
- **Stroke-width perturbation**: morphological dilation (*disk_dilate*) or soft blur (*blur_soft*), radius 1.
- **Affine noise**: rotation $\pm 8°$, scale $\in [0.92, 1.08]$ per axis.
- **Additive noise**: Gaussian, std $\in [0.01, 0.03]$.

With 256 repeats per class, each epoch presents 4,352 synthetically varied examples. Figure 2 illustrates the six augmentation modes.

B. Models

CNN (SimpleCNN). Two convolutional blocks (conv → BN → ReLU → pool) followed by a fully-connected head with 17-way softmax.

Prototype embedding (EmbeddingCNN). The same convolutional backbone projects each image into a 64-dimensional embedding. Prototypes are the mean embeddings of the 17 raw templates. At test time a query is classified by nearest-prototype distance [3].

Both models use Adam (lr $= 10^{-3}$, batch 128) for 8 epochs.

V. Experiments

A. Configurations

Table II lists the five experimental configurations. CNN configurations run for 3 seeds; the prototype model for 5 seeds. Best-epoch MNIST accuracy is reported per seed.

B. Baseline

Nearest-template (L2) [8]: assign each MNIST image the digit label of its closest template under Euclidean distance. No training is required.

TABLE II
EXPERIMENTAL CONFIGURATIONS.

Config	Elastic prob	Stroke prob
no_aug_cnn	0.00	0.00
elastic_only_cnn	0.70	0.00
stroke_only_cnn	0.00	0.80
full_aug_cnn	0.70	0.80
proto (embedding)	0.70	0.80

TABLE III
MNIST TEST ACCURACY (MEAN ± STD %).

Method	Acc. (%)	Seeds
Nearest-template (L2)	38.60	—
No-aug CNN	51.20 ± 1.86	3
Full-aug CNN	65.19 ± 2.14	3
Proto embedding	$\mathbf{77.46 \pm 2.39}$	5
Supervised CNN (reference)	99.11	—

VI. Results

A. Main Results

Table III reports MNIST test-set accuracy for all methods. Figure 3 visualises the comparison.

The prototype-embedding model reaches 77.46%—**78% of supervised CNN performance**—from 17 templates. Even the no-augmentation CNN (51.20%) substantially outperforms the nearest-template baseline (38.60%), showing that training on augmented templates provides significant discriminative capacity beyond simple template matching.

B. Augmentation Ablation

Table IV reports the 2×2 ablation of elastic distortion and stroke-width perturbation.

Both components independently improve accuracy by ~ 5 percentage points over no augmentation, and their combination yields a further gain (∼9 pp over elastic alone). The higher variance of *elastic_only* (±4.44) versus *stroke_only* (±0.78) suggests elastic distortion is more seed-sensitive, while stroke perturbation provides more consistent improvement.

VII. Discussion

Three patterns emerge:

Augmentation is essential. The gap between nearest-template (38.60%) and full-aug CNN (65.19%) confirms that synthetic diversity substantially improves generalisation beyond direct pixel matching.

Embedding beats classification. The prototype model (77.46%) outperforms the classification CNN (65.19%) despite using the same backbone, consistent with findings from Snell et al. [3] that the metric-learning objective is better matched to the few-shot regime.

The ceiling is style coverage, not capacity. Even at 77.46%, the system falls 22 pp short of supervised accuracy. This gap reflects the fact that 17 templates cannot cover all handwriting styles in MNIST regardless of augmentation or model complexity.

TABLE IV
AUGMENTATION ABLATION (CNN, 3 SEEDS, MNIST TEST %).

Augmentation	Elastic	Stroke	Acc. (%)
None	✗	✗	51.20 ± 1.86
Elastic only	✓	✗	56.79 ± 4.44
Stroke only	✗	✓	55.59 ± 0.78
Elastic + stroke (full)	✓	✓	65.19 ± 2.14

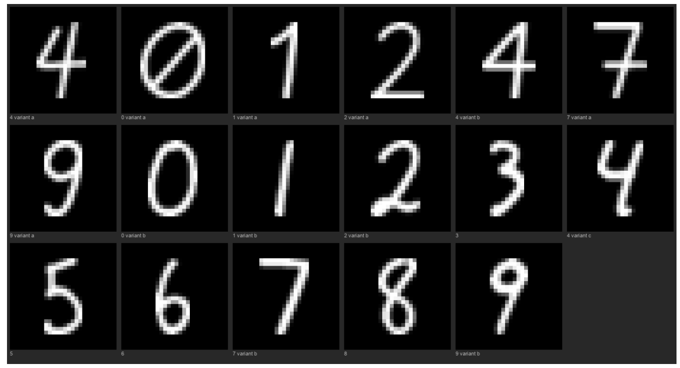

Fig. 1. All 17 CultiVar templates (8× upscaled). Class labels below each digit indicate the style variant.

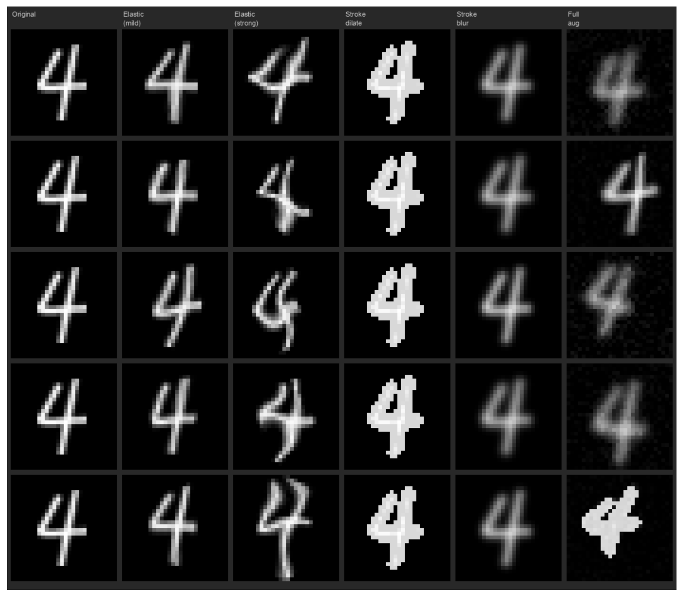

Fig. 2. Augmentation gallery for template 0 (`4_variant_a`). Columns left to right: original, elastic (mild), elastic (strong), stroke dilate, stroke blur, full augmentation (5 samples each).

VIII. THREATS TO VALIDITY

- **Single-source template bias**: all 17 templates originate from one AI-generated composite image and reflect a single stylistic hand.
- **Limited augmentation family**: elastic and stroke perturbations may under-represent real handwriting variability (pen angle, slant, baseline drift).
- **Projection artefacts**: the 17→10 class collapse may hide class-specific errors; a full 17-way confusion matrix would give a more complete picture.

IX. FUTURE WORK

Adding even 2–3 real handwriting samples per style cluster would substantially broaden template coverage. Stronger augmentation (local nonlinear warps, pen-angle simulation) and meta-learning objectives such as MAML [5] or Matching Networks [2] could further improve generalisation. Extending CultiVar-17 to additional cultural conventions and non-Latin scripts is a natural next step for the dataset contribution.

X. REPRODUCIBILITY

All experiments are reproducible with a single command:

```
python scripts/run_oneshot_experiment.py \
       --ablation --proto-seeds 5
```

Figures are regenerated with:

```
python scripts/make_figures.py
```

Raw results are stored in `experiments/reports/oneshot_results.json`.

ACKNOWLEDGEMENTS

[Acknowledgements placeholder.]

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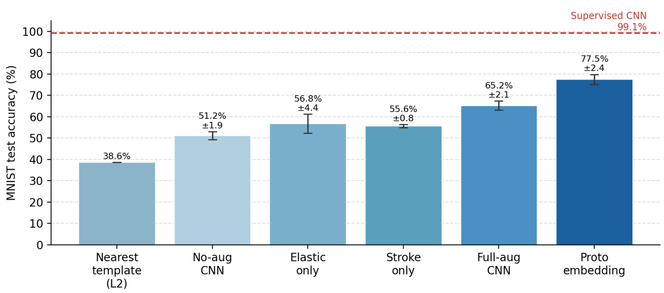

Fig. 3. MNIST test accuracy for all one-shot configurations. Error bars show ± 1 standard deviation across seeds. Dashed line: supervised CNN reference (99.11%).